

RECOMMENDATION ALGORITHMS

Your next *favorite.*

A story about how a feed
learns what to show you.

SAMI / DEPARTMENT OF CSE



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Your next *favorite.*

A story about how a feed
learns what to show you.

Meet Bob.

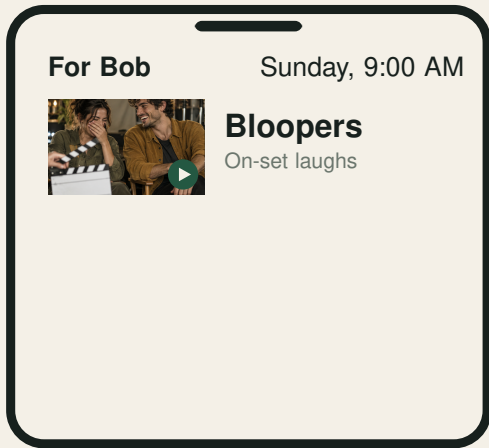
Saturday morning. One funny video.

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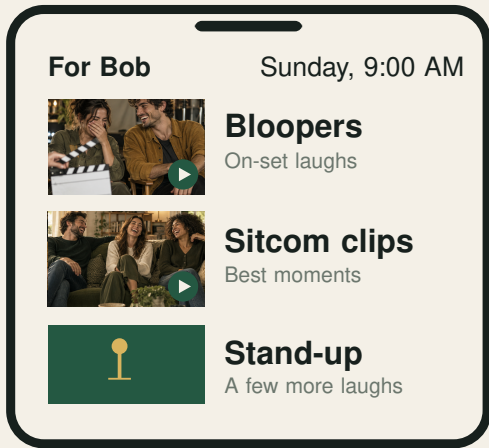
Sunday looks familiar.

One video yesterday.



Sunday looks familiar.

One video yesterday.
A feed full of comedy.

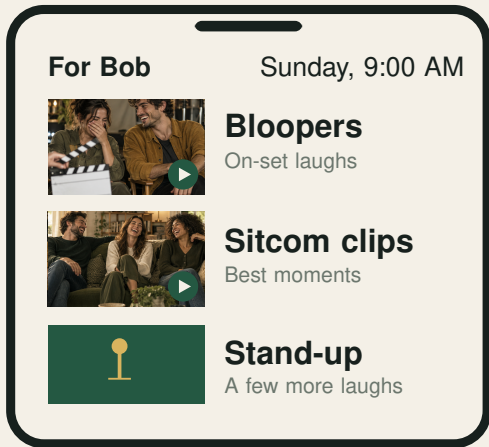


Sunday looks familiar.

One video yesterday.
A feed full of comedy.

Coincidence?

Something is happening
behind the screen.



A ranked guess.

The system uses evidence to decide what Bob might enjoy next.

Observe

Watches,
ratings, skips

A ranked guess.

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Observe

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ratings, skips

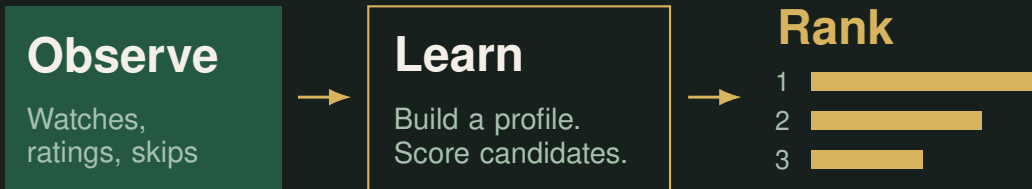


Learn

Build a profile.
Score candidates.

A ranked guess.

The system uses evidence to decide what Bob might enjoy next.



That is the job of a **recommendation system**.

Bob tells us in two ways.

EXPLICIT FEEDBACK

What he says.



Rate. Like. Dislike.

A direct expression of preference.

Bob tells us in two ways.

EXPLICIT FEEDBACK

What he says.



Rate. Like. Dislike.

A direct expression of preference.

IMPLICIT FEEDBACK

What he does.



Watch. Replay. Skip.

Behavior that may suggest interest.

Bob tells us in two ways.

EXPLICIT FEEDBACK

What he says.



Rate. Like. Dislike.

A direct expression of preference.

IMPLICIT FEEDBACK

What he does.



Watch. Replay. Skip.

Behavior that may suggest interest.

A long watch is a clue, not proof that Bob liked it.

A few ratings. Many unknowns.

Rows = people. Columns = videos. Entries = ratings.

	Bloopers	Sitcom 1	Action	Travel
Bob →	5			
Asha				
Leo				

Bob rated
bloopers 5.

A few ratings. Many unknowns.

Rows = people. Columns = videos. Entries = ratings.

	Bloopers	Sitcom 1	Action	Travel
Bob	5	4	?	?
Asha	4	?	2	?
Leo	?	?	5	?

Most ratings
are missing.

A few ratings. Many unknowns.

Rows = people. Columns = videos. Entries = ratings.

	Bloopers	Sitcom 1	Action	Travel
Bob	5	4	?	?
Asha	4	?	2	?
Leo	?	?	5	?

Most ratings
are missing.

Unknown
 $\neq$ disliked.

The challenge: score a video Bob has never rated.

Describe videos. Learn Bob's taste.

Recommend videos whose features resemble what Bob already enjoys.

VIDEOS BOB ENJOYS



Movie bloopers

Describe videos. Learn Bob's taste.

Recommend videos whose features resemble what Bob already enjoys.

VIDEOS BOB ENJOYS



Movie bloopers

ITEM FEATURES

Comedy
Genre, tone, topic,
cast, duration...

Describes the
video.

Describe videos. Learn Bob's taste.

Recommend videos whose features resemble what Bob already enjoys.

VIDEOS BOB ENJOYS



Movie bloopers

ITEM FEATURES

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BOB'S CONTENT PROFILE

$$\mathbf{p} = (3, 1)$$

Comedy: 3 Action: 1

From his feedback
and video features.

A profile of **features**, built from a history of **feedback**.

The closer direction wins.

Compare Bob's profile with candidate video features using cosine similarity.

Action

CANDIDATE VIDEO FEATURES: (COMEDY, ACTION)



A / Sitcom 2

$\mathbf{a} = (4, 0)$



B / Action trailer

$\mathbf{b} = (1, 4)$

Bob (3, 1)

Comedy

The closer direction wins.

Compare Bob's profile with candidate video features using cosine similarity.

Action

CANDIDATE VIDEO FEATURES: (COMEDY, ACTION)



A / Sitcom 2

$\mathbf{a} = (4, 0)$

0.95



B / Action trailer

$\mathbf{b} = (1, 4)$

Bob (3, 1)

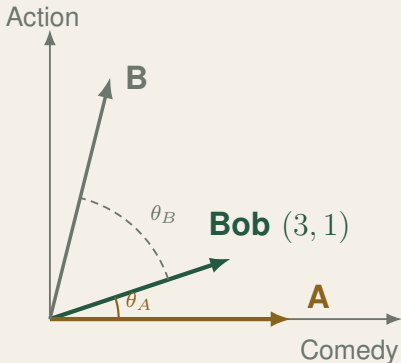
θ_A

A

Comedy

The closer direction wins.

Compare Bob's profile with candidate video features using cosine similarity.



CANDIDATE VIDEO FEATURES: (COMEDY, ACTION)



A / Sitcom 2

$a = (4, 0)$

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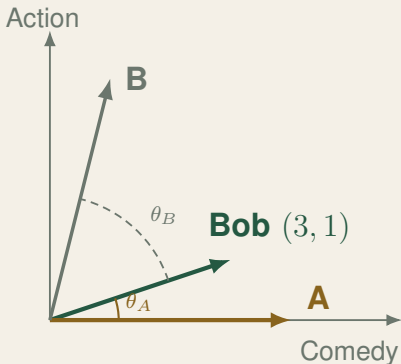
B / Action trailer

$b = (1, 4)$

0.54

The closer direction wins.

Compare Bob's profile with candidate video features using cosine similarity.



CANDIDATE VIDEO FEATURES: (COMEDY, ACTION)



A / Sitcom 2

$\mathbf{a} = (4, 0)$

0.95



B / Action trailer

$\mathbf{b} = (1, 4)$

0.54

$$\text{sim}(\mathbf{p}, \mathbf{v}) = \frac{\mathbf{p} \cdot \mathbf{v}}{\|\mathbf{p}\| \|\mathbf{v}\|}$$

$0.95 > 0.54$

Recommend A.

Can Bob discover *something different?*

Familiar features can mean more of the same.

| What if people with Bob's taste love
a video with completely different tags?

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Collaborative filtering

My teammate takes the story from here.

